Residual TFT-EWC: Continual Learning for State-of-Health Forecasting and Reuse Assessment of Heterogeneous Second-Life EV Batteries

Tarun Sitaraman
Department of Networking and Communications
SRM Institute of Science and Technology
Chennai, India

Sohan Sandeep Krishnan
Department of Networking and Communications
SRM Institute of Science and Technology
Chennai, India

Darshan D
Department of Networking and Communications
SRM Institute of Science and Technology
Chennai, India

***Abstract*—Retired electric-vehicle batteries can serve in second-life applications, but a health model trained on one battery population may fail on the next; continual learning promises to adapt without forgetting. We test this promise for forecasting the state of health (SOH) of lithium-ion cells one to six capacity checks ahead, on a leakage-controlled benchmark from five public datasets (395 cells), with follow-up experiments pre-registered and every claimed difference replicated on a second seed set. A Temporal Fusion Transformer with Elastic Weight Consolidation (TFT + EWC) is compared with fine-tuning, replay, joint training, literature backbones and simple forecasters. Forgetting appears whenever usage conditions differ (0.18–0.46 percentage points, pp). Trained on a shrunk correction to linear extrapolation, the proposed Residual TFT-EWC matches extrapolation on the continual stream and beats it by 0.12–0.20 pp on unseen usage conditions, where gradient boosting is 0.39–0.60 pp more accurate and joint training beats the continual form by 0.32–0.44 pp; it assigns the measured reuse class to 95.5% of unseen-condition samples (gradient boosting 97.7%). Shift-adaptive EWC, first-life pretraining and Spark-extracted ageing-cycle inputs do not help. Conformal intervals under-cover on unseen conditions (0.66–0.76 at 90%) unless calibrated on conditions held out from fitting (0.90), except outside the training range; refitting and tracking lower the 90% interval score by 29%.**

***Index Terms*—Continual learning, second-life batteries, electric vehicles, state of health, battery degradation, heterogeneous batteries, catastrophic forgetting, machine learning**

I. Introduction

When an electric-vehicle (EV) battery no longer meets automotive requirements, it may still serve less demanding applications such as stationary storage or backup power. Reuse extends the battery's life and delays recycling, but only if its health is assessed reliably.

The state of health (SOH) describes the present capability of a battery relative to a reference condition, and machine-learning models have estimated it from voltage, current, temperature, partial charging and impedance measurements [1]–[4]. Retired batteries are, however, heterogeneous: chemistry, manufacturer, capacity, first-life history, temperature and charging protocol differ, and complete first-life records are often missing. A model trained once on a fixed population may therefore fail on the next population. Continual learning addresses this by updating a model on new data while preserving what it learned earlier, so it must be evaluated on both the new and the earlier groups to measure catastrophic forgetting.

This work combines continual learning, health assessment and second-life reuse assessment. Health is assessed by *forecasting* the SOH one to six capacity checks ahead from everything measured up to the latest check: on the first dataset studied, estimating the current SOH from cycling features was beaten by carrying the last measured SOH forward (§III), whereas a reuse decision needs the future SOH. The forecast is mapped to reuse-suitability classes.

The contributions are (Fig. 1):

- A leakage-controlled benchmark for forecasting the SOH of second-life cells from five public datasets (395 cells after quality rules), with two label artefacts removed: test currents set from the previous capacity, and pulse charge counted in the capacity.
- An evaluation of a Temporal Fusion Transformer with Elastic Weight Consolidation (TFT + EWC) against fine-tuning, replay, joint training, simple forecasters and the model families of the literature (BiLSTM, TCN, Transformer, GPR, random forest, SVR) under three protocols (a continual stream, unseen groups and first-life pretraining), reporting adaptation, forgetting and retention.
- A continual TFT + EWC that learns a shrunk correction to linear extrapolation: on the continual stream it matches extrapolation, which no model beats, and EWC reduces its forgetting over longer horizons; on unseen usage conditions it beats extrapolation but not gradient boosting or its jointly trained form (all replicated). Shift-adaptive EWC and first-life pretraining add nothing.
- Forecast intervals for unseen usage conditions: split-conformal intervals calibrated on the training conditions under-cover, and calibrating on conditions held out from fitting restores nominal coverage over the unseen conditions, except for a condition outside the training range.
- A reproducibility analysis: several significance results on

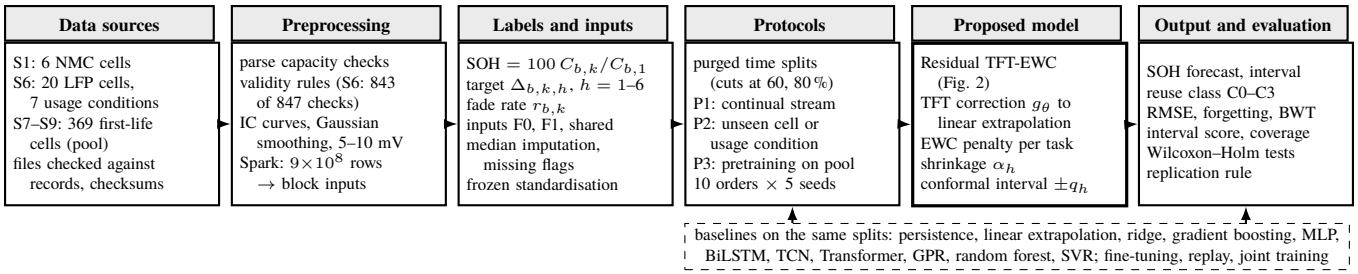


Fig. 1. System architecture: from the public archives to the forecast, its interval and the reuse class, with the protocols and baselines used for evaluation.

TABLE I
APPROACHES OF THE REVIEWED STUDIES AND OF THIS WORK. THEIR ERRORS ARE NOT COMPARABLE WITH OURS (OTHER CELLS, TARGETS AND SOH DEFINITIONS).

Study	Approach	Predicts	CL	Reuse
[1]	partial-charge features	current SOH, second life	–	–
[2]	58 health indicators	current SOH, second life	–	–
[3]	impedance (EIS) + ML	current SOH, second life	–	–
[4]	statistical features, bagged GPR	current SOH, screening	–	✓
[6]	online ML, clustering	current health, second life	–	–
[7]	bilevel-optimised CL	capacity degradation	✓	–
[8]	PNN + TCN	state of charge	✓	–
[9]	domain generalisation	current SOH, second life	–	–
[10]	BiLSTM, TCN, TCN-GPR	SOH with uncertainty	–	✓
This work	residual TFT + EWC, shrinkage, conformal intervals	SOH 1–6 checks ahead, interval	✓	✓

small battery datasets depend on the numerical environment; the main results satisfy a replication rule.

II. Related Work

The review covers 2022–2026 studies (Table I; review in [5]). Second-life SOH has been estimated from partial charging under residential and fast-charging profiles (RMSE about 0.6 and 0.8%, 85% less testing time than a capacity test) [1], from 58 health indicators of reused Nissan Leaf modules (about 0.2% RMSE, but the most useful indicators changed with ageing) [2], from impedance spectra [3], and with bagged Gaussian process regression for screening (worst-case error below 1.48%) [4]; retired Leaf cells aged for over 15 months were tracked online with clustering (error below 2.3%) [6]. These studies estimate the *current* SOH of a fixed population; none adapts a model as new populations arrive. Continual learning has predicted capacity degradation [7] and, with progressive networks and TCNs, the state of charge across drive cycles, addressing forgetting and task order [8]; domain generalisation [9] and probabilistic BiLSTM/TCN models, whose calibration error isotonic recalibration reduced [10], target second-life SOH but not a long sequence of tasks with explicit retention.

Research gap. Second-life SOH estimation, screening, transfer learning and degradation prediction have progressed largely separately; continual learning is rarely combined with heterogeneous second-life SOH forecasting, an explicit reuse decision and an evaluation of forgetting. This work asks whether a continual learner can learn heterogeneous second-life populations in sequence while retaining earlier ones, and whether its forecasts support reuse decisions.

TABLE II
DATASETS. S1 AND S6 FORM THE CONTINUAL-LEARNING STREAMS; S7–S9 FORM THE FIRST-LIFE PRETRAINING POOL.

Set	Cells and conditions	Label	Samples
S1 [11], [12]	6 NMC 21700 cells, second life on residential or commercial peak shaving, 20/35 °C blocks	C/20 RPT capacity, pulses excluded	279
S6 [13], [14]	20 LFP 18650 cells, 7 conditions (charge mode, voltage and rate, discharge, depth), 50 °C	C/25 check capacity	2,409
S7 [15], [16]	195 of 228 NMC/C+SiO cells, calendar, cyclic and WLTP ageing, 0–40 °C	check-up capacity	10,251
S8 [17], [18]	55 NCM cells, six charge/discharge protocols	capacity cycles	9,699
S9 [19], [20]	119 of 130 NCA, NCM and blended cells, 25–45 °C	capacity cycles	16,761

III. Methodology

Given everything measured up to a reference performance test (RPT, or capacity check), we forecast the SOH one to three checks ahead (up to six in follow-up experiments); cells or usage conditions arrive as a stream of tasks. The study first aimed to estimate the *current* SOH from cycle-level features, but on the first dataset these carried almost no health information (median within-cell Spearman correlation with SOH 0.09, against -0.99 for elapsed time) and learned estimators were worse than persistence (1.78 and 2.37 pp for gradient boosting and ridge, against 0.99 pp). We therefore reframed the task after seeing the data, committing the forecasting protocol before fitting any forecasting model.

A. Datasets

Table II lists the five public datasets (all CC BY 4.0). Every file is verified against its repository record or checksums, and our parsing of the raw files reproduces exactly the processed S1 RPT capacities and the authors' capacities of all 843 S6 checks.

S1 contains six LG Chem INR21700-M50T cells that spent 23 months under an EV driving profile [21] and then about 27 months on peak-shaving duty cycles [22]; each RPT is a C/20 discharge interrupted by C/2 pulses, with impedance spectroscopy (EIS) from the fifth RPT [11]. Data-quality findings are handled by fixed rules (e.g. RPT 2, run at about 20 °C, is dropped).

S6 was selected by criteria fixed before screening (licence, a capacity-check series, ageing past first-life end of life, heterogeneity that S1 lacks). Its cells were checked every 100 cycles and end at 32–64% SOH; "second life" means continued ageing below 80% SOH under the same protocol. A check is used only if its C/25 discharge starts from full charge and reaches the cut-off (843 of 847). From the first-life sets *S7–S9*, cells with at least five valid checks are kept (369 cells, 36,711 samples).

B. Big-data processing of the ageing cycles

Between checks, every S6 cell logged each ageing cycle at about one sample per second (827 files, 132 GB, 9.0×10^8 rows). Apache Spark 4.2 (PySpark, local mode) reduces them to the mean, last value and slope of seven measured responses per 100-cycle block (21 inputs), joined to the check that ends the block; the job, its validation against an independent pandas implementation and the excluded quantities are described in the supplement.

C. State of health and forecasting target

For cell b with check capacities $C_{b,1}, C_{b,2}, \ldots$ in time order,

$$\mathrm{SOH}_{b,k} = 100\, C_{b,k}/C_{b,1}\ \%, \tag{1}$$

relative to the first valid measurement, not a nominal rating. In S1 the capacity is the charge of the C/20 steps only [11, Eq. (1)]; the processed tables also counted the pulse charge, shifting some labels by up to 0.09 pp.

A sample is a triple (cell b, origin k, horizon $h \in \{1, 2, 3\}$), about one to three months in S1 and 100–300 cycles in S6. The target is

$$\Delta_{b,k,h} = \mathrm{SOH}_{b,k+h} - \mathrm{SOH}_{b,k}, \tag{2}$$

so an error in Δ equals the error in the future SOH; errors are root-mean-square errors (RMSE) in SOH percentage points (pp). The fade rate $r_{b,k}$ is the least-squares slope of SOH against time t (days or cycles) over the last three valid checks,

$$r_{b,k} = \frac{\sum_{i=k-2}^{k} (t_{b,i} - \bar{t})(\mathrm{SOH}_{b,i} - \overline{\mathrm{SOH}})}{\sum_{i=k-2}^{k} (t_{b,i} - \bar{t})^2}. \tag{3}$$

Inputs use only information available at the origin. **F0** contains the SOH history ($\mathrm{SOH}_{b,k}$, time or cycles since the first check, $r_{b,k}$), static descriptors (S1: duty cycle and first-life history [11]; S6: test conditions) and the *planned* usage until the target (S1: days at each temperature set point and at rest; S6: cycles), so forecasts are conditional on the planned usage. **F1** adds diagnostics measured at the origin: 47 in S1 (pulse resistances, incremental-capacity and EIS features) and 17 in S6 (plateau charges, incremental-capacity peaks, hysteresis, a resistance proxy). No test current is an input: in S1 the RPT and monthly currents were set from the previous capacity and so encode the label. Missing values are imputed from training data and flagged.

D. Splits and protocols

Purged chronological splits. For each cell, cut A lies at 60% and cut B at 80% of its valid checks. Training samples have their target at or before A, validation samples their origin at or after A and target at or before B, and test samples their origin at or after B; samples straddling a cut are dropped, so no training label lies after a validation origin and no validation label after a test origin.

P1, continual stream. A task is a cell in S1 (six tasks) or a test condition in S6 (seven tasks). At step i of an ordering π, a strategy receives only task π_i's data. Each method runs on 10 orders with 5 seeds, identical across methods.

P2, unseen group. One cell (S1) or condition (S6) is held out for testing; models are trained on the others.

P3, first-life pretraining. All datasets are mapped onto shared inputs (SOH history, calendar and cycle time, planned time to the target, operating conditions and chemistry; undefined conditions flagged as missing). Models are pretrained on the pool, which also gives the input scaling and the first EWC anchor; the seven S6 conditions then follow as a stream.

E. Models

Preprocessing (standardisation, median imputation with missing indicators, target scaling) is fitted on the first task and then frozen. The *MLP* has hidden layers of 128 and 64 ReLU units (dropout 0.1). The proposed backbone adapts the Temporal Fusion Transformer (TFT) [23]: a window of the $W{=}4$ most recent origins of the same cell and horizon (inputs only, never labels) passes through per-variable embeddings, variable selection by gated residual networks, an LSTM encoder [24] and interpretable two-head attention masked to past steps (Fig. 2; $d{=}8$, 12,708 parameters with the shared inputs).

Strategies. Fine-tuning trains on the current task only (lower bound); experience replay [25] stores B samples per finished task; joint training fits all tasks at once (upper bound); and Elastic Weight Consolidation (EWC) [26] adds, for each finished task j,

$$\frac{\lambda}{2} \sum_{p} F_{j,p} \left(\theta_p - \theta^*_{j,p}\right)^2, \tag{4}$$

with θ^*_j the parameters after task j and F_j the empirical diagonal Fisher information over up to 500 training samples. *TFT+EWC* is the proposed continual learner; trained on the residual (5) below, with shrinkage and intervals, it forms the proposed *Residual TFT-EWC* (Fig. 2). Training uses Adam [27], batches of 64, at most 200 epochs and early stopping on the current task's validation RMSE.

Reference forecasters are persistence ($\hat{\Delta} = 0$), linear extrapolation $\hat{\Delta}_{b,k,h} = r_{b,k}(t_{b,k+h} - t_{b,k})$, ridge regression (on planned usage only, F0 or F1) and histogram gradient boosting on F1.

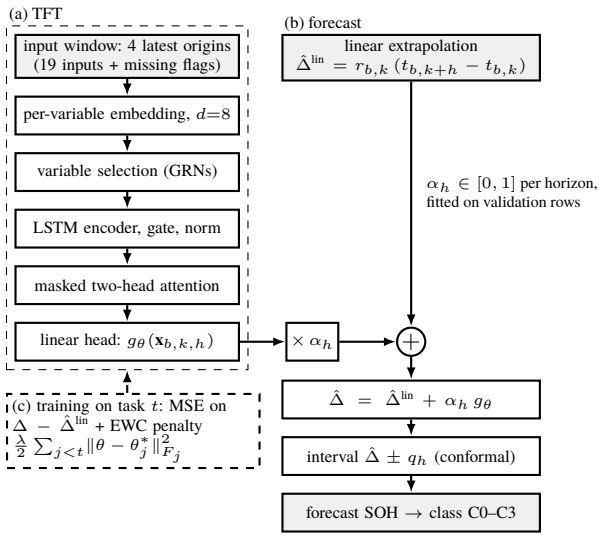


Fig. 2. Proposed model, Residual TFT-EWC. (a) The TFT maps a window of the four latest origins of a cell to a correction g_θ. (b) The correction is shrunk per horizon and added to linear extrapolation (5); conformal intervals give the reuse class. (c) Each task is trained on the correction with the EWC penalty (4).

Learned residual. Because linear extrapolation proved hardest to beat, a pre-registered follow-up trains the continual learners on the correction to it. With $\hat{\Delta}^{\text{lin}}_{b,k,h}$ the extrapolation above (zero when $r_{b,k}$ is missing), the model g_θ learns $\Delta_{b,k,h} - \hat{\Delta}^{\text{lin}}_{b,k,h}$ and the forecast is

$$\hat{\Delta}_{b,k,h} = \hat{\Delta}^{\text{lin}}_{b,k,h} + g_\theta(\mathbf{x}_{b,k,h}). \tag{5}$$

The residual and the direct (target Δ) forms are compared on the S6 stream with the P3 shared inputs, from scratch, for horizons 1–3 (E1) and 1–6 (E2).

Shift-adaptive EWC and literature backbones. A second pre-registered experiment lets the EWC strength follow the shift between groups (shift-adaptive EWC, SA-EWC): when task t starts, the penalty (4) of each finished task j is multiplied by $d_{j,t} = \big((\|\mu_t-\mu_j\|^2 + \|\sigma_t-\sigma_j\|^2)/D\big)^{1/2}$, the per-dimension 2-Wasserstein distance between diagonal Gaussians fitted to the two tasks' preprocessed training inputs, with λ_0 tuned as λ. BiLSTM [10], TCN [8], [10] and Transformer-encoder continual learners (TFT input window, within 20% of its parameters) and single-fit GPR [3], [4], random forest and SVR are run in residual and direct form.

Shrinkage and forecast intervals. A third pre-registered experiment calibrates the residual model after training, leaving its weights unchanged. Per horizon h, the learned correction is shrunk towards extrapolation by $\alpha_h = \min\big(1, \max(0, \sum p\,y / \sum p^2)\big)$, with p the predicted and y the observed correction on calibration rows. A split-conformal interval [28] $\hat{\Delta}\pm q_h$ takes as q_h the $\lceil (n+1)(1-a)\rceil$-th smallest of the n absolute calibration residuals at horizon h. Intervals at level $1-a$ (90% and 80%) are scored by coverage, width and the interval score [29] $2q_h + \frac{2}{a}\max(|\Delta-\hat{\Delta}| - q_h, 0)$ (pp; lower is better). On the stream both are refitted after each task on its validation rows; extrapolation gets intervals the same way.

Unseen conditions. A fourth experiment repeats P2 on S6 for the shrunk residual TFT + EWC, streamed over the six training conditions in the ten evaluation orders with the P1 tuned values; shrinkage and intervals are fitted once on their pooled validation rows. Origins with a fade rate (H^+) and without (H^-, a cell's first two checks, where extrapolation is persistence) are reported separately. A fifth experiment calibrates intervals leaving one condition out: each training condition is predicted by a model fitted on the other five, and the absolute residuals at all its origins, pooled over the six, give q_h (cross-conformal [30], optionally per horizon and stratum). Its forecaster is gradient boosting, the most accurate in P2; being deterministic, it must replicate across both horizon sets instead of seed sets. In a sixth the held-out condition arrives over time: as each target is measured, gradient boosting is refitted with it and the half-width moves by $c\,q_h(e-\alpha)$, $e=1$ for a miss (quantile tracking [31]; c chosen on the training conditions).

F. Hyperparameters, metrics and statistics

Hyperparameters are chosen on validation data from tuning orders disjoint from the evaluation orders: the learning rate per backbone ($\{3\cdot10^{-4},\dots,10^{-1}\}$; lowest validation plasticity of fine-tuning streams), then replay $B\in\{50,\dots,400\}$ and EWC $\lambda\in\{10^0,\dots,10^5\}$; a grid whose best value is at an edge is extended by two values. From P3 on, ten tuning orders and the one-standard-error rule (the smallest grid value within one standard error of the best) are used.

After each step i, every task's test set is evaluated, giving $R_{i,j} = \text{RMSE}(\mathcal{D}^{\text{te}}_{\pi_j} \mid \theta_i)$. Following [32], [33], we report the final average error $\frac{1}{T}\sum_j R_{T,j}$, backward transfer $\frac{1}{T-1}\sum_{j<T}(R_{T,j}-R_{j,j})$ and forgetting $\frac{1}{T-1}\sum_{j<T}(R_{T,j} - \min_{j\le i<T} R_{i,j})$ (positive values mean forgetting), the skill $1-\text{RMSE}/\text{RMSE}_{\text{persistence}}$, and the agreement of the reuse class of the forecast SOH with the measured class (C0 $>95\%$, C1 90–95%, C2 80–90%, C3 $\le 80\%$, fixed in advance).

For P1 and P2 the same predictions are also scored with further regression metrics (MAE, median error, bias, R^2_Δ of the change since the origin against its mean, which is ≤ 0 for persistence, and others) and, on all held-out samples pooled, reuse-class metrics (accuracy, macro-F1, the Matthews correlation coefficient, MCC, and others) and the AUC of the decision SOH $\le 80\%$ (end of first life); the full sets are in the repository (decision D30). Means carry 95% bootstrap intervals [34]. Methods are compared by two-sided Wilcoxon signed-rank tests [35] on matched runs or held-out groups, with rank-biserial effect sizes [36] and Holm correction [37]. A difference d is *replicated* only if it is significant in the main seed set (0–4) and in a replicate set (5–9), with the same sign and $|d| \ge 0.05$ pp in both; $|d| < 0.05$ pp in both means practically equivalent. The threshold is about half the label shift caused by the pulse-charge definition.

G. Implementation and availability

The pipeline uses Python 3.11, NumPy 2.4, pandas 3.0, scikit-learn 1.9 and PyTorch 2.14.1 (P1 and P2 were first run on an Intel Xeon with an unrecorded PyTorch version and re-run on an Intel Core 5 210H), with one CPU thread and seeded, deterministic algorithms. Every result file records configuration, seeds, library versions and the Git commit; unit tests cover input causality, split purging and label exclusion from the inputs. The datasets are at doi:10.17605/OSF.IO/8JNR5 (S1), 10.57745/OLBXKT (S6), 10.35097/1969 (S7), 10.5281/zenodo.10963339 (S8) and 10.5281/zenodo.6405084 (S9); code, decision log, pre-registrations and results are at https://github.com/TarunSitaraman/residual-tft-ewc.

IV. Results

Data. S1 diagnostics track the current SOH (median within-cell $\rho = 0.97$ for the main incremental-capacity peak) but not the change to the next RPT (median $|\rho| \leq 0.3$); every S1 sample stays in class C0, while all S6 cells cross from C0 to C3.

A. Continual stream (P1)

S1. TFT + EWC reaches 0.462 pp (skill 0.27), level with gradient boosting, TFT fine-tuning and MLP replay (0.466–0.475 pp; Table III); a ridge regression on the planned usage alone is slightly better (0.450 pp). Forgetting is small for every method (0.016–0.040 pp). Differences whose significance changed in a re-run are only 0.005–0.014 pp (§IV-B): on S1 the methods are practically equivalent.

S6. When tasks differ in usage, forgetting is substantial (0.18–0.46 pp). EWC lowers the TFT's error and forgetting relative to fine-tuning (0.677 against 0.883 pp), but which difference is significant changed between runs (§IV-B). MLP fine-tuning and MLP + EWC (0.531 and 0.540 pp) are significantly better in both runs, the joint upper bounds (0.414 and 0.427 pp) put the TFT's continual-learning cost at about 0.25 pp, and linear extrapolation (0.343 pp) and gradient boosting (0.372 pp) are the best forecasters.

B. Unseen cell or usage condition (P2)

Gradient boosting is best on both datasets (Table IV). On S6 the 100% depth-of-discharge condition is the hardest for every method (1.8–2.4 pp). With six or seven held-out groups, no difference with the TFT survives Holm correction.

The further metrics (D30, from a re-run within 0.0052 pp of Table IV; full sets in the supplement) give the same ranking: gradient boosting has the largest R^2_Δ (0.54 on S1, 0.74 on S6), while linear extrapolation forecasts the change no better than its mean (-1.31 and -0.05; persistence -1.85 and -3.16).

Reproducibility. A re-run of P1 and P2 in a pinned environment on data rebuilt from the raw archives reproduced every baseline, every P2 result and the ranking of methods; the neural models moved by at most 0.059 pp, but the significance of differences of 0.005–0.014 pp changed (details in the supplement). This motivated the ten tuning orders, the one-standard-error rule and the replication rule used from P3 on: neural training is deterministic per platform, not across platforms.

C. First-life pretraining (P3)

Pretraining on the first-life pool (36,711 samples, against 2,409 in S6) does not help: for the pretrained TFT, EWC and fine-tuning are practically equivalent, both backbones are better from scratch on average (TFT 0.401/0.415 pp against 0.509/0.591 pp pretrained with EWC, significant only in the replicate set), and linear extrapolation (0.343 pp) remains the most accurate method, on S6 and on the pool's own test split (details in the supplement).

Why the TFT improved. A post hoc ablation, repeated on the replicate seeds, explains why the TFT from scratch improves from 0.891 to 0.401 pp with the shared inputs: dropping the 17 diagnostic features (0.17–0.35 pp) and adding the calendar time between checks (0.19–0.31 pp), both replicated ($p_{\text{Holm}} \leq 0.002$).

D. Learned residual on linear extrapolation

For the residual form (5) (Fig. 3; Table SIII of the supplement), the pre-registered criterion, residual TFT + EWC beating linear extrapolation, was *not* met: the two are practically equivalent for horizons 1–3 (differences of -0.004 and -0.029 pp), and for horizons 1–6 extrapolation is better on average ($+0.076$ and $+0.049$ pp, not replicated). Per horizon, extrapolation is better at $h = 1$ and 2 (replicated), and the gap closes without reversing by $h = 5$–6. Learning the residual does make the TFT competitive: it improves TFT + EWC over the direct form by 0.17 pp (E1) and 0.24–0.30 pp (E2) and puts it ahead of gradient boosting, all replicated. For horizons 1–6, where it forgets more, EWC lowers its forgetting from 0.28/0.22 to 0.09 pp and its backward transfer from 0.18/0.14 to 0.03 pp, replicated; its effect on the final error is not. For horizons 1–3, EWC and fine-tuning are practically equivalent. *Ageing-cycle inputs.* The 21 Spark-extracted inputs (Section III-B), added as pre-specified, make the residual TFT + EWC worse: 0.480/0.501 against 0.339/0.314 pp, and worse than linear extrapolation, both replicated.

E. Shift-adaptive EWC and literature baselines

The second pre-registered criterion, for SA-EWC (Table SIII of the supplement), was *not* met: the residual TFT with SA-EWC and with fixed λ are practically equivalent in final error and forgetting for both horizon sets (differences of at most 0.03 pp), although the shift weights varied (0.05–1.14). For horizons 1–3, BiLSTM + EWC (direct, 0.323 pp, and residual) and the residual Transformer + EWC are practically equivalent to the residual TFT; for 1–6 it beats the residual BiLSTM and TCN (replicated; not the Transformer). Every single-fit regressor is worse than the residual TFT + SA-EWC (replicated), except the residual SVR for horizons 1–6.

TABLE III

PROTOCOL P1 (CONTINUAL STREAM). FINAL AVERAGE TEST RMSE OF SOH(k+h) IN PP (MEAN ± S.D. OVER 10 ORDERS × 5 SEEDS FOR CONTINUAL METHODS, OVER 5 SEEDS FOR JOINT TRAINING), SKILL RELATIVE TO PERSISTENCE ($1 - \mathrm{RMSE}/\mathrm{RMSE}_{\mathrm{persistence}}$) AND FORGETTING (PP). †UPPER BOUND, TRAINED ON ALL TASKS AT ONCE.

Method	S1: six NMC second-life cells RMSE	Skill	Forgetting	S6: 20 LFP cells, seven usage conditions RMSE	Skill	Forgetting
Persistence	0.632	0.00	–	1.644	0.00	–
Linear extrapolation	0.726	-0.15	–	0.343	0.79	–
Ridge, usage only	0.450	0.29	–	1.274	0.22	–
Ridge, F0	0.678	-0.07	–	0.801	0.51	–
Ridge, F1	0.491	0.22	–	0.605	0.63	–
Gradient boosting, F1	0.466	0.26	–	0.372	0.77	–
MLP fine-tuning	0.492 ± 0.029	0.22	0.030	0.531 ± 0.151	0.68	0.207
MLP replay	0.475 ± 0.021	0.25	0.025	0.597 ± 0.178	0.64	0.212
MLP + EWC	0.497 ± 0.033	0.21	0.040	0.540 ± 0.176	0.67	0.212
MLP joint†	0.500 ± 0.016	0.21	–	0.414 ± 0.037	0.75	–
TFT fine-tuning	0.467 ± 0.014	0.26	0.026	0.883 ± 0.538	0.46	0.461
TFT + EWC	0.462 ± 0.019	0.27	0.016	0.677 ± 0.261	0.59	0.274
TFT + EWC, F0	0.472 ± 0.020	0.25	0.019	0.621 ± 0.257	0.62	0.176
TFT joint†	0.455 ± 0.017	0.28	–	0.427 ± 0.049	0.74	–

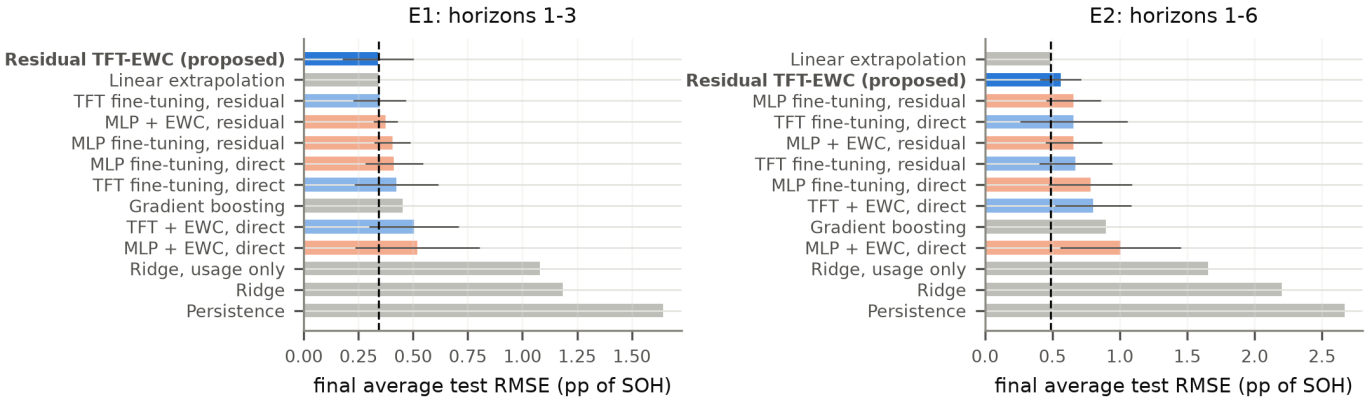


Fig. 3. S6 continual stream: final average test RMSE (pp), mean and s.d. over the 50 runs of the main seed set; bold, blue: the proposed residual TFT-EWC; dashed: linear extrapolation. Replicate set, forgetting and literature backbones: Table SIII of the supplement.

F. Shrinkage and intervals on the stream

Shrinking the correction per horizon makes the residual TFT + EWC practically equivalent to extrapolation at every horizon (0.318/0.308 against 0.343 pp for horizons 1–3, 0.498/0.493 against 0.485 pp for 1–6), removing its replicated loss at $h = 1$–2; the correction is turned down most at $h = 1$ (Fig. S7 of the supplement). The criterion was again not met. Its 90% intervals are narrower than extrapolation's but undercover (0.71–0.81, against 0.92–0.95), so the interval scores are equal (practically equivalent or not replicated). Pretraining the residual model on the first-life pool makes it worse for horizons 1–3 (by 0.22 pp, replicated) and does not help for 1–6.

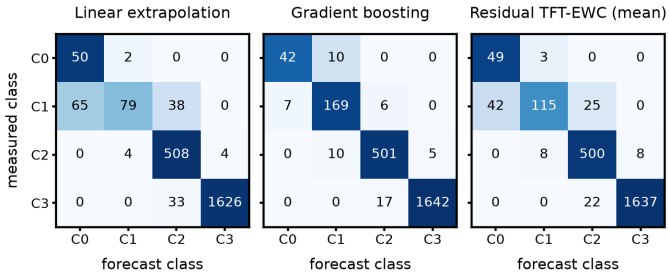


Fig. 4. Reuse-class confusion on unseen S6 conditions (Table VI): sample counts (proposed model: mean over 50 runs); shading: share of the measured class.

G. Residual model on unseen conditions

In the fourth experiment (Table V), as registered, the shrunk residual TFT + EWC beats extrapolation at origins with a fade rate, by 0.12/0.12 pp for horizons 1–3 and 0.20/0.18 pp for 1–6 (replicated): the first comparison in which the proposed model beats extrapolation. Elsewhere extrapolation is persistence and loses by over 2 pp. Gradient boosting, however, beats the TFT by 0.39/0.41 pp and 0.57/0.60 pp (replicated), on six of the seven conditions; only on the 100% depth-of-discharge condition is the TFT better. The same residual TFT trained jointly on the six conditions beats its continual form by 0.32–0.44 pp (replicated). Shrinkage and the raw correction are practically equivalent here.

The reuse classes follow the same ranking (Table VI, Fig. 4).

TABLE IV

PROTOCOL P2 (UNSEEN BATTERY IN S1, UNSEEN USAGE CONDITION IN S6): MEAN TEST RMSE (PP) OVER HELD-OUT GROUPS, RANGE OVER GROUPS, AND FOR S6 REUSE-CLASS ACCURACY / MACRO-F1 OF THE FORECAST SOH.

Data	Method	RMSE	Range	Class acc./F1
S1	Persistence	0.545	0.436–0.688	–
S1	Linear extrapolation	0.511	0.284–0.725	–
S1	Ridge, usage only	0.301	0.172–0.409	–
S1	Ridge, F0	0.418	0.158–0.779	–
S1	Ridge, F1	0.320	0.161–0.477	–
S1	Gradient boosting, F1	0.219	0.091–0.299	–
S1	MLP	0.285	0.149–0.389	–
S1	TFT	0.292	0.199–0.377	–
S6	Persistence	2.817	2.025–4.327	0.85 / 0.64
S6	Linear extrapolation	1.372	0.856–1.669	0.94 / 0.75
S6	Ridge, usage only	1.177	0.870–2.288	0.95 / 0.83
S6	Ridge, F0	0.859	0.598–1.836	0.96 / 0.86
S6	Ridge, F1	1.137	0.588–2.188	0.96 / 0.85
S6	Gradient boosting, F1	0.710	0.395–1.839	0.97 / 0.88
S6	MLP	0.821	0.476–2.225	0.96 / 0.87
S6	TFT	0.868	0.551–2.390	0.96 / 0.88

TABLE V

UNSEEN S6 USAGE CONDITION (P2, SEVEN HELD-OUT CONDITIONS): TEST RMSE (PP), MEAN OVER FOLDS, ON ALL ORIGINS AND ON ORIGINS WITH A FADE RATE (H^+). TFT ROWS: MEAN OVER 50 RUNS, MAIN / REPLICATE SEED SETS; OTHER ROWS: SINGLE DETERMINISTIC FITS.

	E1: h=1–3		E2: h=1–6	
Method	all	H^+	all	H^+
TFT + EWC, residual, shrunk	0.92 / 0.94	0.74 / 0.74	1.62 / 1.65	1.42 / 1.44
TFT, residual, joint	0.57 / 0.58	0.54 / 0.55	1.21 / 1.18	1.22 / 1.19
Gradient boosting	**0.54**	**0.51**	**1.05**	**1.05**
Ridge	0.77	0.69	1.30	1.26
Linear extrapolation	1.37	0.86	2.30	1.62
Persistence	2.82	2.66	4.90	4.73

The proposed model assigns the measured class to 95.5% of held-out samples (94.4–96.8% over the 50 runs), against 93.9% for extrapolation and 85.8% for persistence, but ridge (96.5%) and gradient boosting (97.7%) do better. All errors are between neighbouring classes; most forecast a measured C1 sample one class healthier (C0): 65 of 182 for extrapolation, 42 for the proposed model and 7 for gradient boosting.

Intervals on unseen conditions. Intervals calibrated on validation rows of the training conditions under-cover on the held-out one (pooled 0.66–0.68 for the TFT, 0.74–0.76 for extrapolation); the calibration rows, not only the model, cause the shortfall. With leave-one-condition-out calibration, gradient-boosting intervals reach the registered 0.85 (0.90) with the lowest interval scores (3.2 and 6.7 pp), but not on the 100% depth-of-discharge condition, outside the range of the others (0.48 and 0.38; details and Fig. S8 in the supplement).

Refit and tracking. Refit and tracking lower the mean 90% interval score by 29%, from 3.22 to 2.29 pp (h=1–3; six of seven conditions, $p = 0.047$) and from 6.67 to 4.74 pp (h=1–6; seven of seven, $p = 0.016$), with pooled coverage 0.90–0.91. On the 100% depth-of-discharge condition coverage rises from 0.48/0.38 to 0.72/0.58, short of the registered 0.80. With these intervals, 87–93% of held-out forecasts lie within one reuse class, 99.2–99.7% of them correctly.

TABLE VI

REUSE CLASS OF THE FORECAST SOH ON UNSEEN S6 CONDITIONS (P2, HORIZONS 1–3, SEVEN HELD-OUT CONDITIONS POOLED; MEASURED C0 52, C1 182, C2 516, C3 1659). SENS., PREC.: MACRO RECALL AND PRECISION. PROPOSED MODEL: MEAN AND S.D. OVER 50 RUNS; OTHERS: SINGLE FITS. BOLD: BEST.

Approach	Acc.	Sens.	Prec.	F1	MCC
Persistence	0.858	0.782	0.625	0.661	0.723
Linear extrapolation	0.939	0.840	0.810	0.777	0.876
Ridge	0.965	0.857	0.861	0.858	0.927
Gradient boosting	**0.977**	**0.924**	**0.926**	**0.925**	**0.952**
Residual TFT-EWC (proposed)	0.955	0.880	0.840	0.840	0.906
s.d. over 50 runs	±0.005	±0.017	±0.016	±0.021	±0.011

V. DISCUSSION

Why the proposed model does not win. On the stream TFT + EWC never beat linear extrapolation: fade over one to three checks is smooth, the groups are few and small (21–27 and 135–225 training samples per task in S1 and S6), and the TFT's attention over its history is nearly uniform (supplement). EWC helps where there is something to protect (S6, not S1), most clearly on the residual target, where it cuts forgetting over six horizons by about two thirds; a shift-adaptive penalty added nothing, and at horizons 1–3 a BiLSTM or Transformer with EWC is as accurate as the residual TFT. On unseen conditions the continual constraint itself is the main cost (joint training is 0.32–0.44 pp more accurate): with data this small, storing every earlier group is cheap, so the constraint that motivates continual learning did not bind.

What matters. Building on the best simple forecaster mattered most, and the ablation points to inputs, not architecture: diagnostic features hurt the TFT and the calendar time between checks helped, consistent with fade per cycle rising with that time at 50 °C in all 20 S6 cells. For intervals, the calibration rows mattered as much as the model: only residuals from held-out conditions reproduce the error on a new one.

Reproducibility and reuse. As significance changed across platforms (Section IV-B), we recommend recording the environment, tuning over enough runs and requiring replicated differences above a practical threshold. The reuse classes are informative only when the horizon crosses class boundaries, and for a population unlike any seen, such as the 100% depth-of-discharge condition, neither forecasts nor intervals can be trusted, which pooled coverage hides.

Limitations. The task was changed to forecasting after the first dataset was inspected; the second-life datasets are small; S6 is aged at 50 °C and its "second life" continues the same protocol; SOH is relative to each cell's first valid measurement; forecasts are conditional on planned usage; and the ablation and every later experiment were decided after the main results (all except the ablation were committed before they were run). The single-fit regressors see all tasks at once, and the direct literature backbones were run for horizons 1–3 only. The TFT hyperparameters used on unseen conditions were tuned with validation rows of all seven S6 conditions, which come from

one laboratory, with only training randomness varied. Leave-one-condition-out intervals were run for gradient boosting and extrapolation only.

VI. Conclusion

We evaluated a TFT with EWC as a continual learner for forecasting the SOH of second-life cells, on a leakage-controlled, pre-registered benchmark. Learning a shrunk correction to linear extrapolation made it as accurate as extrapolation on the stream and more accurate on unseen usage conditions, where gradient boosting and the same TFT trained jointly were clearly better: at this data scale, learning continually cost more than it gave. Conformal intervals held on unseen conditions only when calibrated on conditions held out from fitting, and not outside the training range; refitting and tracking cut the interval score by 29%, and 87–93% of forecasts fell within one reuse class, over 99% of them correctly. Future work should test continual learning where earlier data cannot be stored and populations are larger, and calibrate intervals across laboratories and chemistries.

References

[1] E. Braco, I. San Martín, P. Sanchis, A. Ursúa, and D.-I. Stroe, "State of health estimation of second-life lithium-ion batteries under real profile operation," *Applied Energy*, vol. 326, p. 119992, 2022.

[2] ——, "Health indicator selection for state of health estimation of second-life lithium-ion batteries under extended ageing," *Journal of Energy Storage*, vol. 55, p. 105366, 2022.

[3] M. Faraji-Niri, M. Rashid, J. Sansom, M. Sheikh, D. Widanage, and J. Marco, "Accelerated state of health estimation of second life lithium-ion batteries via electrochemical impedance spectroscopy tests and machine learning techniques," *Journal of Energy Storage*, vol. 58, p. 106295, 2023.

[4] A. Takahashi, A. Allam, and S. Onori, "Evaluating the feasibility of batteries for second-life applications using machine learning," *iScience*, vol. 26, p. 106547, 2023.

[5] V. S, H. S. Che, J. Selvaraj, K. S. Tey, J. W. Lee, H. Shareef, and R. Errouissi, "State of health (SoH) estimation methods for second life lithium-ion battery—review and challenges," *Applied Energy*, vol. 369, p. 123542, 2024.

[6] X. Cui, M. A. Khan, G. Pozzato, S. Singh, R. Sharma, and S. Onori, "Taking second-life batteries from exhausted to empowered using experiments, data analysis, and health estimation," *Cell Reports Physical Science*, vol. 5, p. 101941, 2024.

[7] M. Lee, S. Kim, S. Kim, and J.-I. Choi, "Bilevel-optimized continual learning for predicting capacity degradation of lithium-ion batteries," *Journal of Energy Storage*, vol. 86, p. 111187, 2024.

[8] J. Yao, B. Zheng, and J. Kowal, "Continual learning for online state of charge estimation across diverse lithium-ion batteries," *Journal of Energy Storage*, vol. 117, p. 116086, 2025.

[9] X. Li, Y. Lin, S. Yang, and W. Zhang, "Intelligent domain-generalized second-life EV battery state-of-health estimation," *Journal of Energy Storage*, vol. 140, p. 118989, 2025.

[10] S. Meskine, S. Al-Majeed, and H. El Asri, "Reliability assessment of second-life EV batteries using probabilistic deep learning models for state-of-health prediction," *World Electric Vehicle Journal*, vol. 17, p. 441, 2026.

[11] K. Moy, M. A. Khan, S. Fasolato, G. Pozzato, A. Allam, and S. Onori, "Second-life lithium-ion battery aging dataset based on grid storage cycling," *Data in Brief*, vol. 57, p. 111046, Dec. 2024.

[12] ——, "Second-life lithium-ion battery aging dataset based on grid storage cycling," Open Science Framework, dataset, file version 2 (27 Feb. 2025), 2024, licensed under CC BY 4.0. Accessed 29 Sep. 2026. [Online]. Available: https://doi.org/10.17605/OSF.IO/8JNR5

[13] W. Wheeler, Y. Bultel, P. Venet, and A. Sari, "An ageing study of twenty 18650 lithium-ion graphite/LFP cells in first and second life use," *Scientific Data*, vol. 12, p. 392, 2025.

[14] W. Wheeler, P. Venet, Y. Bultel, and A. Sari, "Aging study on twenty A123 18650 graphite/LFP 1.1 Ah cells," Recherche Data Gouv, dataset, version 2.0 (11 Mar. 2025), 2025, licensed under CC BY 4.0. Accessed 29 Sep. 2026. [Online]. Available: https://doi.org/10.57745/OLBXKT

[15] M. Luh and T. Blank, "Comprehensive battery aging dataset: capacity and impedance fade measurements of a lithium-ion NMC/C-SiO cell," *Scientific Data*, vol. 11, p. 1004, 2024.

[16] ——, "Comprehensive battery aging dataset: capacity and impedance fade measurements of a lithium-ion NMC/C-SiO cell [dataset – version 2: result data]," Karlsruhe Institute of Technology, RADAR4KIT, 2024, cC BY 4.0.

[17] F. Wang, Z. Zhai, Z. Zhao, Y. Di, and X. Chen, "Physics-informed neural network for lithium-ion battery degradation stable modeling and prognosis," *Nature Communications*, vol. 15, p. 4332, 2024.

[18] F. Wang, "Project – physics-informed neural network for lithium-ion battery degradation stable modeling and prognosis," Zenodo, version V1, 2024, cC BY 4.0.

[19] J. Zhu, Y. Wang, Y. Huang, R. Bhushan Gopaluni, Y. Cao, M. Heere, M. J. Mühlbauer, L. Mereacre, H. Dai, X. Liu, A. Senyshyn, X. Wei, M. Knapp, and H. Ehrenberg, "Data-driven capacity estimation of commercial lithium-ion batteries from voltage relaxation," *Nature Communications*, vol. 13, p. 2261, 2022.

[20] ——, "Data-driven capacity estimation of commercial lithium-ion batteries from voltage relaxation [dataset]," Zenodo, 2022, cC BY 4.0.

[21] G. Pozzato, A. Allam, and S. Onori, "Lithium-ion battery aging dataset based on electric vehicle real-driving profiles," *Data in Brief*, vol. 41, p. 107995, Apr. 2022.

[22] K. Moy and S. Onori, "Synthetic grid storage duty cycles for second-life lithium-ion battery experiments," *SAE International Journal of Advances and Current Practices in Mobility*, vol. 6, no. 1, pp. 261–269, 2023.

[23] B. Lim, S. Ö. Arık, N. Loeff, and T. Pfister, "Temporal fusion transformers for interpretable multi-horizon time series forecasting," *International Journal of Forecasting*, vol. 37, no. 4, pp. 1748–1764, 2021.

[24] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[25] D. Rolnick, A. Ahuja, J. Schwarz, T. P. Lillicrap, and G. Wayne, "Experience replay for continual learning," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[26] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska, D. Hassabis, C. Clopath, D. Kumaran, and R. Hadsell, "Overcoming catastrophic forgetting in neural networks," *Proceedings of the National Academy of Sciences*, vol. 114, no. 13, pp. 3521–3526, 2017.

[27] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *International Conference on Learning Representations (ICLR)*, 2015.

[28] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman, "Distribution-free predictive inference for regression," *Journal of the American Statistical Association*, vol. 113, no. 523, pp. 1094–1111, 2018.

[29] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," *Journal of the American Statistical Association*, vol. 102, no. 477, pp. 359–378, 2007.

[30] R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani, "Predictive inference with the jackknife+," *The Annals of Statistics*, vol. 49, no. 1, pp. 486–507, 2021.

[31] A. N. Angelopoulos, E. J. Candès, and R. J. Tibshirani, "Conformal PID control for time series prediction," in *Advances in Neural Information Processing Systems*, vol. 36, 2023, arXiv:2307.16895.

[32] D. Lopez-Paz and M. Ranzato, "Gradient episodic memory for continual learning," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[33] A. Chaudhry, P. K. Dokania, T. Ajanthan, and P. H. S. Torr, "Riemannian walk for incremental learning: Understanding forgetting and intransigence," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018.

[34] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York: Chapman & Hall, 1993.

[35] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.

[36] D. S. Kerby, "The simple difference formula: An approach to teaching nonparametric correlation," *Comprehensive Psychology*, vol. 3, p. 11.IT.3.1, 2014.

[37] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.